



Constructing Warren Truss Bridge for Solving Urban Congestion

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Abstract

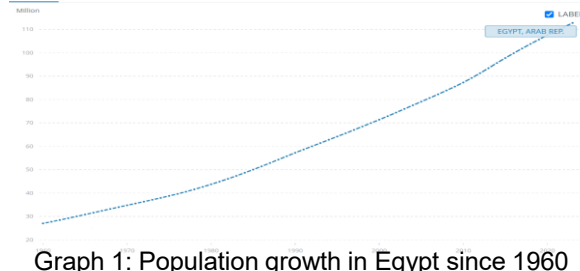
There are eleven major problems that Egypt has suffered throughout years. Today, they are termed challenges. These challenges inflicted their burden on the Egyptians' well-being and made many of their activities very likely impossible. Some of the challenges sequenced in this essay include towns with nearby populations, a very dry place, and climate change-all problems that are interrelated and affect one another. Egypt is on the lookout for different palliatives and solutions over this problem. In this regard, there is a made bridge using the Warren Truss type specifically. With a Warren truss, the force can be distributed throughout the surface of the whole bridge. This means that all the load on the bridge will be evenly distributed along its length. The midstationary part is about 14.5 centimeters clear from the ground level. The movable part is 28 cm long as to requirement design. The bridge is Vertical-lift. It is a highly efficient mode of operating this bridge and very reliable. It has a composition with the help of cables and pulleys, used to raise the deck from the ground, allowing boats to float under it. This kind of bridge is easier to design and erect than other bridge types. The price is comparatively cheaper. Also, it requires less maintenance than other similar bridges. Hence, the total cost is less. The Vertical-lift bridge can be operated in different environmental conditions and used according to the need. Weights ranging from 0.5 kg to 10.2 kg would test the performance of the bridge. The maximum deflection noted was 5 mm under the load of 10.2 kg. With no support from the centre columns when it was hung in mid-air, there was no deflection under dead load, thereby proving the excellence of the trusses and load distribution.

Introduction

Urban congestion has proven to be a heightened and serious problem, and it remains one of the many problems that Egypt has faced for almost 80 years. Such kind of movement should considerate that Egypt does have urban congested cities, which developed mostly during the past ten years and showed no sign of recovery. This continuity was because it is one of the major problems, a problem of overpopulation where more than 70 million people have added the country just in 60 years as depicted in graph 1. Such were the impacts of these problems on the economy of Egypt, on its productivity, on the environment, public health, and general life and well-being of its people. Therefore, this is the reason that the GCMA is now home to more than 19 million foreign residents whose lives have been encumbered within various types of perils of life, which then has proved to greatly hamper contributions to the GDP.



Figure 1: high-tonnage hydraulic cylinder



Graph 1: Population growth in Egypt since 1960

To curb this urban congestion, Egypt has built numerous cities to ensure proximity to the problems of aridity and overcrowding. El Rhoda Island is an example among such many areas. An area with diversified sectors across economic activity with job opportunities, urban congestion remains the biggest drawback to turning its opportunities into reality, thereby creating inefficiencies, a dip in economic fortunes, and health-related issues. This suggests that there is something more to be done beyond that remedy for the gap, like constructing bridges.

The alleviation of congestions is hub specific with building and design specifications. The model for El Rhoda Island is such that it will have two traffic lanes (7 cm each) and two pedestrian lanes (3 cm each) separated by a 2 cm strip at the center, with a clearance of 10 cm from the lowest point to the water's surface. The bridge should be load-bearing on a minimum of 10 kg with an outright deflection of 5 cm and weighs not more than 20 kg. Research on hydraulic and mechanical motion, Newton's laws, and openable bridge designs fed into the building of the prototype.

Among several advantages, hydraulic motion is easy to adapt with different sizes of ships, as fewer materials are involved, thus very cost effective, very durable against heavy traffic, and even better traffic flow due to reduced waiting times. Specialized maintenance creates a high liability for downtime, but then all hydraulic bridges necessitate this. Among different hydraulic bridge models, the best option for El Rhoda Island would have to be the vertical lift bridge. All it needs is a small amount of horizontal space and provides great vertical clearance for vessels passing under it, which makes it most suitable for the 50-meter span proposed, unlike swing bridges.

These hydraulic cylinders (as shown in figure 1) will shift this energy into linear mechanical energy because they are capable of lifting very high loads. Furthermore, such characteristics ensure that the bridge is not only effective in eliminating congestion but increasingly effective in traffic flow conversion and even economic productivity-toward formulation of a very significant step towards addressing the major urban challenges facing Egypt.

Materials & Methods

ITEM	QUANTITY	USAGE	COST	PICTURE
PALM STALK	100 OF STALK	THE BASIC MATERIAL FOR THE CONSTRUCTION OF THE BRIDGE	100 L.E	
WHITE GLUE	2	GLUING MATERIALS TOGETHER	120 L.E	
SYRINGE	4	TO PUMP THE MOVABLE PART OF THE BRIDGE	20 L.E	
HOSE SYRINGE	1	TO LINK WITH THE SYRINGE AND THE MOVABLE PART OF THE BRIDGE	20 L.E	
YELLOW GLUE	1	GLUING MATERIALS TOGETHER	60 L.E	
Total cost			325	

Methods:

The prototype was built of 5 main phases. Firstly, the base structure was built by the usage of the palm stalk which was used to make a rectangular base then the corners were glued with white glue and the base was reinforced with additional stalk. Secondly, the Hinge Mechanism was created through the construction of triangular support at the ends of the base for additional strength. Then cubes were made from palm to create a hinge after that the dowel was cut to size and thread through the cube to allow the bridge to pivot. Thirdly, the bridge platform was constructed by gluing sticks in rectangular pattern to form it then the attachment of the platform to the hinge mechanism was ensured so that it can lift and lower smoothly. Fourthly, the hydraulic system was set up by linking between the syringe and the hose to form the hydraulic system where the input and the output one were acting as switches. Oil was then poured to the syringe to give some pressure. The stable base and the platform were connected to control the lifting mechanism. The last phase was testing the mechanism and modifications were made to ensure smooth lifting.

Test plan:

After building the prototype there should be a test to ensure its effectiveness so, there will be tests done. This test shows if the prototype is reliable or not. Those test mainly revolve around testing the deflection, reliability and other aspects of the prototype. To test the reliability of the bridge, a deflection test will be done. This test will measure if the prototype can carry dead load and live load with little deflection. In this test, the prototype will have its centre hung mid-air to see if there is deflection from the dead load of it. Then, masses will be added on its midpoint to see if it can resist them without collapsing or not. Mass put on the prototype will go from 0.5 kg to 10 kg increasing by 0.5 kg each time. To achieve the design requirements, the deflection of the bridge shouldn't exceed 5 cm.

Results

Mass(KG)	Deflection(mm)
0.0	0.0
0.5	0.0
1.0	0.0
1.5	0.0
2.0	0.0
2.5	0.0
3.0	0.0
3.5	0.0
4.0	0.0
4.5	1.0
5.0	1.0
5.5	1.0
6.0	2.0
6.5	2.0
7.0	4.0
7.5	4.0
8.0	5.0
8.5	5.0
9.0	5.0
10.2	5.0

The prototype had a capability for weight support of a maximum of up to 4 kg without showing any significant or considerable additional deflections at all. Beyond the maximum weight a slight deflection of 1 (+/-0.5) mm was observed when loaded with a weight of 4.5 kg. Furthermore the deflection continued gradually as the weight increased adding 1 (+/-0.5) mm for every several kilograms until it reached a maximum of 7.2 kg. At this point, the amount of deflection increased with an additional load going up 2 (+/-0.5) mm which indicated that the speed of deflection was two times larger than that which had been previously recorded under regular conditions. In fact, the bridge managed to hold over 10.2 kg total weight, much greater than the required mass, under just 5(+/- 0.5) mm of deflections.

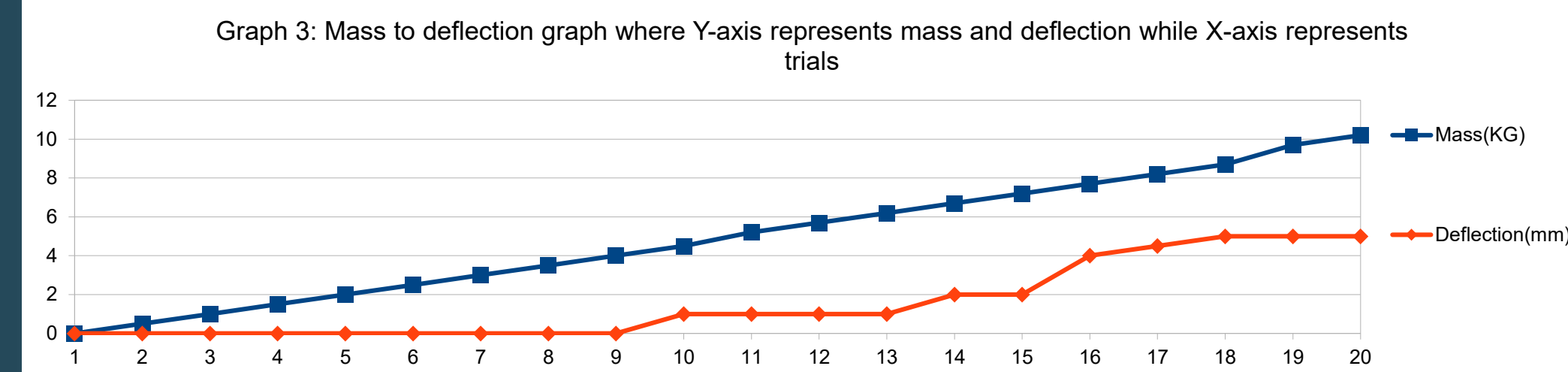
Positive results:

- 1)The bridge was able to take the loads without significant deflections.
- 2)This stress test was passed with 0.2 kg more than what was required for the challenge, in evidence of strength and reliability.
- 3)It further met all design parameters that were part of this challenge easily and efficiently.

Negative results:

- 1) The deflection in the bridge was perceptibly high when weight was applied at a single point.
- 2) There is systematic error of the order of (+/- 0.5) mm which it must be taken in consideration looking at the performance.

Analysis



As shown from the results in the previous section, the prototype showed a variety of positive and negative results. The positive results can be summarized into 4 statements.

The first point is that when mass is increased, the deflection increases at a lesser rate in comparison with the rest. This goes to show how well built it is and how it carries the mass over the trusses and columns in the bridge. This is because of the Warren Truss, which is the truss that is best in distributing the weights under dynamic load conditions. A feature of this truss is that it has diagonals which form a chain of triangles. These triangles help in supporting live and dead loads. In addition, it distributes force over the surfaces of the bridge, which is very important for any bridge, because in this way, force or tension is not able to collapse and fall apart. During the test, these trusses helped to hold the prototype together despite being hanged mid-air and made for very visible results because the weight was distributed throughout every part of the bridge. However, it could definitely carry its dead load because of its own structure and the live load added to it during testing. This happens due to the perfect load transfer provided by the geometrical shape, which is the hallmark of the Warren truss. It maintains strength while not wasting the material on it. The Warren Truss is a truss that handles loads by location so that some members carry compression while other carries tensile loads. The Warren truss is made to optimize for lightweight materials while being cost-effective. The Pressure was calculated during this test by the law.

$$\text{Pressure} = \text{Force/Area}$$

Another reason for involving the Warren truss design in our prototype is that the structure has an outstanding strength-to-weight ratio that could be expressed as how much load the bridge could withstand compared with the weight of the entire bridge itself. The statement may actually be such a huge advantage that one could qualify it as the supreme virtue of an entire truss structure. This is even more true when the total weight of the entire bridge is not more than 20 kilograms. From that amazing characteristic, the bridge can sustain a further additional load of 0.2 kilograms from all other comparative bridges tested in the process. In addition, it was discovered that the foundation along with the trusses could actually carry a weight of about 80 kilograms without much deflection when several alterations were made to the bridge design and structure. Also, another major element influencing the bridge's higher strength-to-weight ratio is rooted in the concepts that **Newton's Third Law of Motion expounds**. This basic law states that **"for every action there is an equal and opposite reaction."** In other words, when a load is born by the bridge, every force applied is countered by an equal and opposite force emanating from the opposite direction. Such laws thus prove, with great clarity and conviction, how such a bridge can hold and support several times greater weight than its structure permit. Oddly enough, this is given that there will be such a great and sizeable disparity between the total weight of the bridge and all additional loads it is meant to carry in safety.

And the final reason for its usage is because it has one of the least deflections. The reasons why things deflect are because of the high span length, which leads to greater deflection. They have a relation that **if Span doubles, the deflection increases by 8 times**. Another thing affecting the deflection is the load type because regularly distributed loads have less deflection than concentrated ones. And that's one of the reasons of the design requirement in the challenge that states that the maximum length of the truss should be 15 cm. This is required because of this relation between span length and deflection.

Basically, the problems that the results highlight are about the case when an irregular loading pattern occurs while the Warren truss supports a load. At that time, some of the members always get subjected to such a high level of stresses, which ultimately results in bending deflection within the structure. This type of construction should be achieved so that, under absolutely proper conditions, weight is uniformly distributed to carry the truss members under idealized conditions. Such perfect construction was, however, never reached during the first trial test of the bridge, despite the fact that several revisions and modifications were possible after testing in order to eliminate the possible errors. The Warren truss is a construction needing specialists who can feel and measure all angles absolutely to be sure that it is constructed perfectly in order to serve the purpose.

Systematic error of (+/-0.5) mm was there in measuring tape, having only 1mm markings, not at every 0.1mm markings for precision and accuracy.

Conclusion

Prototype construction was completed and the test was conducted. The summarized results are as follows:

- 1)The unloaded height from the ground of the bridge is 14.5 cm.
 - 2)The structure showed no deflections during the tests until weights of up to 4.5 kg were applied, at which point it deflected by 1 mm.
 - 3)The bridge is made from eco-friendly materials and does no harm to the environment.
 - 4)At 10.2 kg, the bridge reached a maximum deflection of 5 mm.
 - 5)The maximum dimension of the truss in the bridge did not exceed 15 cm to ensure that it does not exceed 5 mm.
 - 6)A load of 75 kg can be maintained by the bridge.
- These findings led us to conclude that the bridge meets all design expectations and is therefore considered a success.

Recommendations

However, there could be suggestions that can improve the efficiency and make such work easier in the future:

- 1)including new futuristic technologies that can bring new things into the project.
- 2)using more recycled materials to make the bridge eco-friendly.
- 3)putting as much effort towards the stationary part as the movable part since they are equally essential.
- 4)finding out the best adhesive material and buying it to avoid buying another one in the future which would increase the cost of the prototype.

Suggestions for constructing a large-scale bridge include:

- 1) there should be an electricity conducting feature from converting kinetic energy to run mechanisms on it.
- 2) the materials used should be reliable and elastic capable to withstand earthquakes and deflect without breaking.

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